Hyunwoo Yang

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EDUCATION

DigiPen Institute of Technology | Redmond, Washington **May/2028**
B.S. in Computer Science , Real-Time Interactive Simulation| GPA: 3.91
Relevant Courses: Computer Graphics, Operating Systems, Data Structures and Algorithms, Databases, Network Programming

TECHNICAL SKILLS

Programming Languages: C/C++, C#, JavaScript, GLSL, Assembly, MySQL, HTML / CSS
Engines: Custom Engine, Unity, Unreal, Godot
Tools: OpenGL, Vulkan, Linux, RenderDoc, Perforce, Git, Github
Other Skills: Multithreading(job system)

PROJECTS

Graphics Calculator | yanghyunw00.github.io/project/calc **Jun/2026 - Sep/2026**

- Full-stack bilingual (ko/en) math web app built with React 19, Vite, Three.js and Math.js, offering 18 solvers across matrix, calculus, vector and 3D graphics, each returning step-by-step derivations rendered as LaTeX through KaTeX.
- Cut initial JS payload 83% (504 KB to 85 KB gzip) with React.lazy route-level code splitting, deferring Three.js, mathjs and KaTeX to the routes that need them. Deployed as a static build on GitHub Pages.
- Built the computation layer on Math.js and Nerdamer covering determinants, inverses, LU/SVD decomposition, eigenvalues, derivatives, integrals, limits and Taylor series, emitting structured solution steps rather than a bare answer.
- Implemented real-time Three.js visualization: MVP matrix transforms with three light types and shadow mapping (Basic/PCF/PCF Soft), vector cross products and parametric surface meshes for partial derivatives.

Lost Guns | Solo Developer | **DOWNLOAD** **Jan/2026 -Mar/2026**

- Recreated the core mechanics of the mobile game LostGuns as a 2D multiplayer shooter in Unity, built from scratch by studying gameplay footage.
- Reverse-engineered four core systems — movement, jumping, ladder traversal and weapon switching and added friend-to-friend online play for up to 8 players via Unity Relay.
- Integrated 2D assets by rigging them with DragonBones and manually authoring skeletal animations to faithfully replicate the original game's visual dynamics.
- Implemented server-authoritative hit detection with lag compensation, rewinding player positions to account for RTT and interpolation delay, clamped to a 0.4s maximum.

I.G.Y | TECH LEAD | IGY_presskit **Jan/2026 - Mar/2026**

- Designed and implemented A* pathfinding with spatial partitioning to navigate up to 500 citizen agents in real time and applied backtracking to solve constraint-based placement problems in the city simulation.
- Profiled the pathfinder and found 91% of node-reset work was wasted (1.93M nodes cleared to expand 160K).
- Rewrote it with generation-stamped lazy initialization and stack-allocated neighbor buffers, cutting per-node cost 1.6µs → 0.84µs and a measured 179ms frame spike to a projected 111ms.
- Replaced the immediate-mode renderer with a GPU-instanced batch renderer, cutting draw calls 9,256 → 11 and frame time 58.6 → 38.3ms (17 → 26 FPS), by packing per-quad transform, UV and tint data into a 96-byte instance record batched across 64 texture slots.
- Expanded a Computer Graphics coursework renderer into a standalone engine, Dokev, abstracting raw OpenGL into single-function calls so teammates new to graphics could focus on gameplay and AI.
- Led technical architecture and version control workflow for a 5-person team, translating gameplay design goals into an engine roadmap built on custom C++/OpenGL.

LEADERSHIP EXPERIENCE

Military Service | Squad Leader (Sgt.) | Daegu, South Korea **Aug/2025 - Feb/2025**

- Led and trained an 8-person mortar squad, coordinating operations and communicating clearly under pressure to align the team and make time-sensitive decisions together.

AWARDS

- **Departmental Valedictorian** — awarded full-tuition merit scholarship
- **Korea–U.S. Advanced Technology Scholar** — selected for a national program supporting promising science and engineering students, receiving up to $18,000 in funding